

Function-Aware Fill-in-the-Middle as Mid-Training for Coding Agent Foundation Models

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Abstract

1 Coding agents must integrate external tool returns into ongoing reasoning—a capability that standard left-to-right pretraining on code exposes only in its forward
 2 direction. We observe that the *action* $\rightarrow$ *observation* $\rightarrow$ *continuation* loop of
 3 a coding agent is structurally isomorphic to a function call site, where a caller
 4 binds arguments, a callee returns a value computed elsewhere, and downstream
 5 code consumes that value. This conditioning structure exists at internet scale
 6 in ordinary code. We exploit it through *function-aware fill-in-the-middle (FIM)*
 7 *mid-training*: a self-supervised objective that masks functions selected via program
 8 dependency graph analysis and a complexity–inferability double criterion. We mid-
 9 train Qwen2.5-Coder (7B/14B) and Qwen3-8B on a 2.6B-token decontaminated
 10 corpus drawn from 968 GitHub repositories, then apply existing agentic post-
 11 training pipelines (R2E-Gym, SWE-Smith) without modification. Mid-training
 12 improves SWE-Bench-Verified by +2.8/ +3.0 points at 7B/14B and by a compar-
 13 able +3.2 on Qwen3-8B, and the improvement holds across post-training pipelines
 14 and base-model families. Beyond in-domain gains, mid-training also mitigates
 15 the capability erosion that agentic post-training otherwise inflicts on non-agent
 16 coding (e.g., LiveCodeBench) and non-coding tool-use benchmarks (tau-bench,
 17 BFCL): although the mid-training corpus contains Python code only, the function-
 18 call inductive bias survives post-training and yields consistent improvements over
 19 post-training alone.
 20

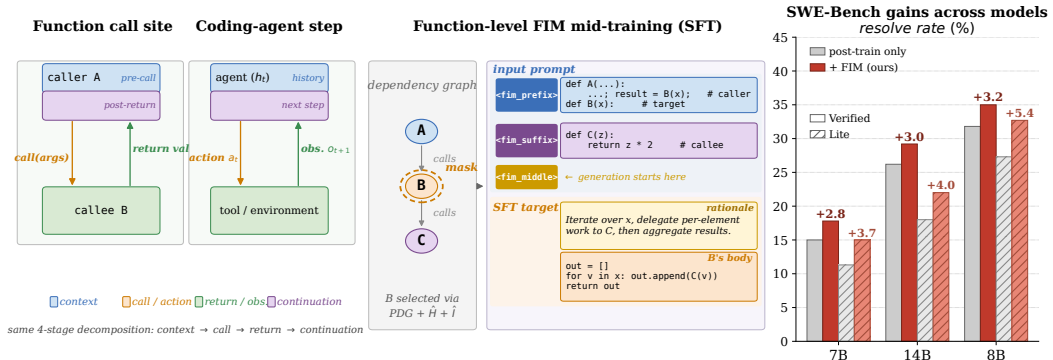


Figure 1: **Left:** A function call site and a single step of a coding agent are *structurally similar*, decomposing into the same four stages: context, call/action, return/observation, continuation. **Middle:** We exploit this analogy via function-aware FIM mid-training. A function B is selected from the program dependency graph using complexity ($\hat{H}$) and inferability ($\hat{I}$) scores; the model is then fine-tuned to produce a chain-of-thought rationale followed by B 's body, given the surrounding file as an FIM-formatted prompt. **Right:** Mid-training yields consistent gains across both Qwen2.5-Coder (7B, 14B) and Qwen3 (8B) on SWE-Bench-Verified (solid bars) and SWE-Bench-Lite (hatched bars).

21 1 Introduction

22 Coding agents that resolve real software engineering issues have moved from research demos to
 23 deployed systems [10, 21, 18]. Their progress, however, has been driven almost entirely by scaling
 24 synthetic agent-trajectory data during post-training: pipelines such as SWE-Gym [14], R2E-Gym [9],
 25 and SWE-Smith [22] curate or synthesize trajectories that imitate human or LLM behavior on
 26 issue-resolution tasks. The base model these pipelines start from is typically a code LLM trained
 27 with next-token prediction (and, in some cases, random-span FIM) on internet-scale code [7, 5, 11].
 28 Between these two stages lies a training-time gap: the base model is rarely optimized for the
 29 conditioning structure that agentic post-training will later demand. We treat this gap as an opportunity
 30 for a dedicated *mid-training* stage that aligns the base model with agent-relevant inductive biases
 31 before agent-specific data is introduced.

32 Our central observation is that the inductive bias required by a coding agent already exists in ordinary
 33 code, but in a shape that left-to-right pretraining systematically under-exposes. At each step an agent
 34 maintains a history h_t , samples an action $a_t \sim \pi(a_t | h_t)$, receives an observation o_{t+1} produced by
 35 an external process, and continues conditioned on the entire trace. This four-part decomposition—
 36 context, action, externally-computed return, continuation—is precisely the decomposition of a
 37 function call site: pre-call code that establishes intent and binds arguments; the call itself; a return
 38 value produced by code outside the immediate scope; and downstream code that consumes the return
 39 value (Figure 1, left). A model trained to reason *bidirectionally* about function-level dependencies
 40 must learn to reconstruct a callee’s behavior from caller context and downstream usage, which is the
 41 same competence required to predict an agent’s continuation given a history and a tool return. The
 42 correspondence is structural rather than literal—FIM training conditions on a given suffix while agent
 43 inference generates one—but it suggests that representations induced by the former should transfer to
 44 the latter, an empirical question we address in Section 3.

45 The fill-in-the-middle objective itself is not new, dating back to early work on code LLMs [1];
 46 recent models mix random-span FIM into their pretraining corpora [7, 5, 11]. We argue, however,
 47 that random-span FIM is poorly aligned with the agentic conditioning structure for three reasons.
 48 (i) *Span boundaries are syntactically arbitrary*: most masked spans cut through expressions or
 49 partial statements and provide only weak signal about function-level dependencies. (ii) *There is no*
 50 *reasoning supervision*: the model fills the span directly, with no intermediate rationale mirroring
 51 an agent’s think-then-act pattern. (iii) *The objective is dissolved into pretraining*: by the time agent
 52 post-training begins, any structural prior conferred by FIM has been amortized across trillions of
 53 unrelated tokens. We address all three points. We select masking targets at function granularity using
 54 program dependency graph analysis combined with a complexity–inferability double criterion that
 55 is deliberately base-model-agnostic (Section 2.3). We embed chain-of-thought rationales *inside* the
 56 FIM middle span so that the model learns to produce a reasoning trace consistent with the eventual
 57 code (Section 2.4). And we apply the objective at a dedicated mid-training stage, concentrating its
 58 signal immediately before agentic post-training rather than spreading it across pretraining.

59 We evaluate this recipe along three axes of robustness. *Across model sizes*, mid-training improves
 60 SWE-Bench-Verified by +2.8 and +3.0 points at 7B and 14B respectively: the gain is consistent
 61 across the scales we evaluate, indicating that the structural prior is not absorbed by larger pretrained
 62 models and remains useful in the practical deployment range. *Across post-training pipelines*, mid-
 63 training improves both R2E-Gym and SWE-Smith on the same 7B base, with the SWE-Smith pairing
 64 yielding +5.3 points on SWE-Bench-Verified. *Across base-model families*, mid-training transfers
 65 from Qwen2.5-Coder to Qwen3-8B with a +3.2-point gain on SWE-Bench-Verified (paired with the
 66 SWE-Lego post-training pipeline), suggesting the prior is not specific to a single pretraining recipe.

67 A second set of results probes our motivating hypothesis. We note first that all benchmarks in
 68 this paper are evaluated on checkpoints that have undergone the full pipeline—either post-training
 69 alone (baseline) or mid-training followed by post-training (ours)—because a mid-trained checkpoint
 70 without subsequent instruction-style post-training has degraded instruction-following ability and is
 71 not directly comparable to instruction-tuned baselines. Under this protocol, we observe that agentic
 72 post-training alone substantially regresses non-target capabilities at 14B: it drops LiveCodeBench,
 73 BFCL, and tau-bench by double-digit margins in some cases. Adding mid-training before the same
 74 post-training largely restores these capabilities, recovering +11.1 points on LiveCodeBench and
 75 gaining +2.4 on BFCL and +3.9 on tau-bench over post-training alone. Because the mid-training
 76 corpus contains only Python code with no tool-use data, mid-training a priori has no reason to affect

tau-bench or BFCL; the fact that it does, and in the same direction as the in-domain SWE-Bench gains, provides direct evidence for the isomorphism between function calls and agent tool calls.

Contributions. (1) **Conceptual framing.** We articulate a structural correspondence between function call sites and the action–observation–continuation loop of coding agents, and motivate function-granularity FIM as a self-supervised prior for agent capability that exists at internet scale in ordinary code. (2) **Method.** We introduce a function-aware FIM mid-training pipeline combining program dependency graph analysis, a hand-designed complexity–inferability double criterion that is base-model-agnostic, and chain-of-thought rationales embedded inside the FIM middle span. (3) **Three-axis robustness.** We empirically validate the recipe along three independent axes—model size, post-training pipeline, and base-model family—observing consistent in-domain gains in every configuration. (4) **Direct test of the motivating hypothesis.** The same Python-only mid-training corpus—which contains no tool-use data—also yields gains on tau-bench, BFCL, and LiveCodeBench, providing direct evidence that the function-call inductive bias survives agentic post-training and transfers across task families. (5) **Open release.** We release the 968-repository decontaminated corpus (78K filtered self-contained Python files, $\approx$ 33M lines), the full selection pipeline, and all mid-training checkpoints to support reproduction.

2 Method

2.1 Motivation: Function Calls as Agent-Like Structures

At step t a coding agent maintains a history h_t , samples an action $a_t \sim \pi(a_t \mid h_t)$, receives an observation $o_{t+1} \sim p(o_{t+1} \mid h_t, a_t)$, and continues conditioned on the trace, $\pi(\cdot \mid h_t, a_t, o_{t+1})$. Under the function-call/agent-step correspondence (Figure 1, left), h_t aligns with the pre-call context, a_t with the call, o_{t+1} with the return, and the continuation with downstream usage of the return value. This motivates extracting an agentic conditioning signal from the structure already present in source code via a fill-in-the-middle (FIM) objective. The correspondence is *structural* rather than literal—FIM conditions on a given suffix while agent inference generates one—but a model trained to reason bidirectionally about function-level dependencies should learn a representation that transfers when the suffix is generated rather than observed. Standard random-span FIM [7, 5, 11] captures this structure only incidentally; we construct a *function-aware* variant that selects masking targets based on program structure and contextual predictability.

2.2 Data Collection and Decontamination

We curate a corpus of 968 Python repositories from GitHub. Starting from $\sim$ 2,000 candidates retrieved by combining a star-count threshold with ten topic queries, we apply manual quality filtering and remove every repository whose origin overlaps with the source repositories of SWE-Bench [10] (verified by repository name and any known fork). To eliminate test-time leakage, we restrict each repository to commits whose timestamp precedes the earliest base-commit used in SWE-Bench-Verified and SWE-Bench-Lite. The filtered corpus contains $\approx$ 78K self-contained Python files (averaging 428 lines per file) and yields $\sim$ 2.6B training tokens under the Qwen2.5-Coder tokenizer, spanning 320K single-function, 60K pair, and 20K triple FIM samples; full statistics, topic-category breakdown, and license inventory are reported in Appendix A, and the full repository list will be released to support reproduction.

2.3 Function-Aware FIM Target Selection

Given each source file, our pipeline produces a set of mask targets together with the corresponding masked file. Each target is a single function or a connected group of 2–3 functions; the multi-function variant is studied in Section 3.4. The pipeline has four stages: dependency-graph construction, complexity scoring, inferability scoring, and threshold-based selection. Figure 2 illustrates these stages on a small running example—a Calculator class with two top-level helpers. The full single-function selection algorithm is given in Appendix B.1, the multi-function extension in Appendix B.7, and a numerical walkthrough of every quantity shown in Figure 2(b) in Appendix B.6.

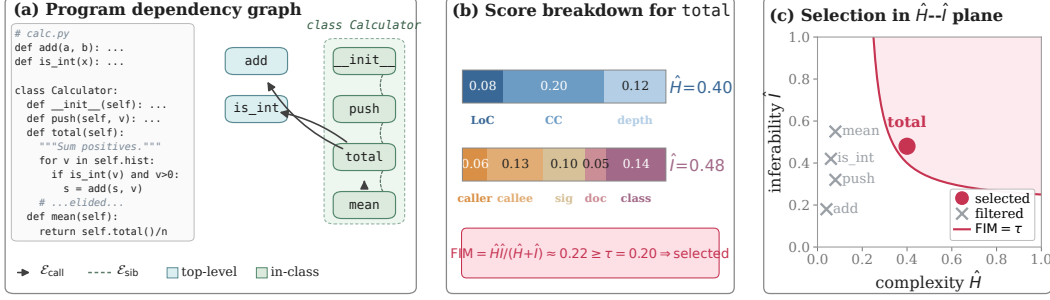


Figure 2: Function-aware FIM target selection on a small calculator example. **(a)** Program dependency graph parsed from the AST: solid arrows are call edges $\mathcal{E}_{\text{call}}$, dashed lines are sibling edges $\mathcal{E}_{\text{sib}}$ between same-class methods. **(b)** Stacked bars decompose the complexity score $\hat{H} = 0.40$ (Eq. 1; LoC, CC, depth) and the inferability score $\hat{I} = 0.48$ (Eq. 2; five context signals) for `Calculator.total`, yielding $\text{FIM} \approx 0.22 \geq \tau$. **(c)** In the $(\hat{H}, \hat{I})$ plane, the four short helpers fall outside the selected region while `Calculator.total` is kept.

2.3.1 Program Dependency Graph

For each file we parse its AST and extract the set $\mathcal{V}$ of function nodes (top-level functions and class methods, identified by qualified names). Two edge sets are constructed: *call edges* $\mathcal{E}_{\text{call}}$ between caller and callee, and *sibling edges* $\mathcal{E}_{\text{sib}}$ between methods of the same class (capturing intra-class coupling that does not manifest as direct calls but carries information through shared instance state). Call resolution handles common Python idioms (direct invocations, class instantiation, `self/ccls` method calls) with a short-name fallback against the qualified-name index; Appendix B.2 gives the full procedure.

2.3.2 Complexity Score $\hat{H}$

For each $v \in \mathcal{V}$ we define

$$\hat{H}(v) = w_\ell \phi(\text{LoC}(v), c_\ell) + w_c \phi(\text{CC}(v), c_c) + w_d \phi(\text{D}(v), c_d), \quad (1)$$

where $\text{LoC}(v)$ is lines of code, $\text{CC}(v)$ is McCabe cyclomatic complexity, $\text{D}(v)$ is the maximum nesting depth of control-flow constructs, and $\phi(x, c) = \min(x/c, 2)$ normalizes each quantity by a soft cap. Caps and weights are listed in Appendix B.3.

2.3.3 Inferability Score $\hat{I}$

A target should be *recoverable* from the surrounding context. $\hat{I}(v)$ aggregates five context-derived signals that approximate the mutual information between v 's body and the rest of the file:

$$\hat{I}(v) = \alpha C_{\text{caller}}(v) + \beta C_{\text{callee}}(v) + \gamma C_{\text{sig}}(v) + \delta C_{\text{doc}}(v) + \varepsilon C_{\text{class}}(v). \quad (2)$$

C_{caller} scores call-site argument specificity, C_{callee} counts intra-file functions called by v , C_{sig} aggregates type annotations and name descriptiveness, C_{doc} indicates docstring presence, and C_{class} counts in-class siblings; full component formulas and weights are in Appendix B.4. Each component is a hand-designed proxy: a learned predictability score would couple selection to a particular reference model and complicate the cross-base-model generalization analysis (Section 3.2).

2.3.4 Single-Function Score

We combine $\hat{H}$ and $\hat{I}$ in a harmonic-mean-like form, scaled by a one-sided difficulty penalty $\rho(\Delta(v))$:

$$\text{FIM}(v) = \frac{\hat{H}(v) \hat{I}(v)}{\hat{H}(v) + \hat{I}(v) + \epsilon} \cdot \rho(\Delta(v)). \quad (3)$$

The harmonic-mean form forces both $\hat{H}$ and $\hat{I}$ to be large simultaneously, penalizing imbalance; ρ down-weights targets that remain hard even given full context, which would otherwise be unlearnable noise. Hard filters on length and dunder methods, the form of ρ , and the threshold $\tau = 0.20$ used throughout this work are documented in Appendix B.5.

2.3.5 Multi-Function Group Selection

Real-world code patches frequently span multiple related functions [10], motivating an extension that masks *groups* of $k=2$ or $k=3$ structurally connected functions; we use this variant in our main recipe and ablate it in Section 3.4. A group score $\text{FIM}(G)$ multiplies a coupling term, the harmonic-mean-like product of group-level $\hat{H}(G)$ and $\hat{I}(G)$, and a difficulty penalty, with $\hat{I}(G)$ recomputed under joint masking so that intra-group references cannot inflate the score. The eight topology patterns (caller-callee, co-callee, sibling-coupled, mutual-call, call-chain, hub, fan-in, class-triad), the full equations, Algorithm 2, and a worked pair example are in Appendix B.7.

2.4 Chain-of-Thought Augmentation

For each selected target we generate a natural-language rationale via Gemini-3-Preview prompted with the masked file and the original function body (template in Appendix B.9). The rationale is embedded *inside* the FIM middle span, immediately before the function body:

```
<fim_prefix> <prefix> <fim_suffix> <suffix> <fim_middle> <rationale> <body>
```

The model is therefore trained to produce, in order, a reasoning trace followed by code consistent with that trace, mirroring the think-then-act structure of an agent step. To control quality, each (CoT, completion) pair is then scored by Gemini-3-Flash given the masked file, the rationale, the generated completion, and the ground-truth function body; we retain only the top-scoring $\sim 400\text{K}$ samples as our final training corpus. The verification prompt template is in Appendix B.9. Section 3.4 isolates the contribution of FIM structure from CoT distillation via a *self-CoT* variant in which the model under training generates its own rationales.

3 Experiments

3.1 Setup

Benchmarks. We evaluate on three groups. *Coding-agent benchmarks:* SWE-Bench-Verified (500) and SWE-Bench-Lite (300) [10], our primary in-domain target. *Non-agent coding benchmarks:* LiveCodeBench [8], OJBench [20], and FullStackBench-EN [2]; these probe pure code generation and serve as a regression check. *Tool-use and OOD agent benchmarks:* Terminal-Bench [12], τ -bench [23], and BFCL [15]; the latter two contain no Python code-editing trajectories and test whether the function-call inductive bias transfers across task families.

Mid-training and post-training pipeline. We mid-train Qwen2.5-Coder-Instruct (7B/14B) and Qwen3-8B on the selected FIM corpus using the standard FIM loss on the middle span (rationale plus body), packed to the model’s native context length and using its native FIM sentinel tokens. We then apply existing agentic post-training pipelines without modification: R2E-Gym [9] or SWE-Smith [22] for the Qwen2.5-Coder runs, and SWE-Lego [17] for Qwen3-8B. For Qwen3-8B we train SWE-Lego for 2 epochs rather than the official 4 to prevent overfitting in our setup. Hyperparameters and token budgets are in Appendix C.

Evaluation protocol. All numerical results are means over three independent evaluation seeds on the final checkpoint of each pipeline; the “%” symbol is omitted inside table cells. Standard-deviation bands shown in red ($\pm?$) are placeholders awaiting final replication runs. “Baseline” refers to the base model followed by an agentic post-training pipeline; “ours” refers to the same base followed by FIM mid-training and then by the identical post-training pipeline. We do not evaluate mid-training-only checkpoints, because FIM data without subsequent instruction-style post-training degrades instruction-following and makes any comparison against instruction-tuned baselines unfair. Every gain we attribute to mid-training is therefore one that *survives* subsequent post-training. The agent harness is fixed by the post-training pipeline (R2E-Gym uses an OpenHands fork [18, 9]; SWE-Smith uses SWE-agent [21]; the Qwen3-8B runs use OpenHands directly because SWE-Lego post-training data exceeds the 32K context of the R2E-Gym/SWE-Smith scaffolds).

Models and baselines. For Qwen2.5-Coder-Instruct at 7B/14B [7] we report (i) the instruction-tuned base, (ii) post-training with R2E-Gym reproduced under our setup, (iii) the official R2E-Gym numbers (in grey, where available), (iv) our recipe of FIM mid-training followed by R2E-Gym post-training, and at 7B additionally the analogous SWE-Smith [22] variants. Qwen3-8B uses the SWE-Lego [17]

Table 1: Main results on coding agent benchmarks. All numbers are percentages, means over three independent seeds with std bands; *Average* is the unweighted mean of Verified and Lite. Bolded rows are our recipe; rows tagged *officially reported* are quoted from the corresponding original publications. SWE-Lego is post-trained for 2 epochs (vs. 4 in the official release) to prevent overfitting.

Setting	SWE-Bench-Verified	SWE-Bench-Lite	Average
<i>Qwen2.5-Coder-7B-Instruct</i>			
— (no agentic training)	1.8 (± 1.3)	1.0 (± 1.0)	1.40
+ R2E-Gym [9] (reproduced)	15.00 (± 1.5)	11.33 (± 1.2)	13.17
+ R2E-Gym (officially reported)	19.00 (± 1.0)	11.00 (± 0.8)	15.00
+ FIM-Midtrain + R2E-Gym	17.80 (± 1.4)	15.00 (± 1.1)	16.40
Δ (ours vs. reproduced)	+2.80	+3.67	+3.24
+ SWE-Smith [22] (reproduced)	12.30 (± 1.2)	14.20 (± 1.4)	13.25
+ SWE-Smith (officially reported)	15.20	11.70	13.45
+ FIM-Midtrain + SWE-Smith	17.60 (± 1.3)	14.70 (± 1.0)	16.15
Δ (ours vs. reproduced)	+5.30	+0.50	+2.90
<i>Qwen2.5-Coder-14B-Instruct</i>			
— (no agentic training)	4.0 (± 1.6)	2.7 (± 1.0)	3.35
+ R2E-Gym (reproduced)	26.20 (± 1.4)	18.00 (± 1.1)	22.10
+ R2E-Gym (officially reported)	26.80 (± 1.4)	20.67 (± 0.7)	23.74
+ FIM-Midtrain + R2E-Gym	29.20 (± 1.5)	22.00 (± 1.2)	25.60
Δ (ours vs. reproduced)	+3.00	+4.00	+3.50
<i>Qwen3-8B</i>			
— (no agentic training)	7.60 (± 1.2)	5.80 (± 0.9)	6.70
+ SWE-Lego [17] (reproduced)	31.80 (± 1.0)	27.30 (± 1.1)	29.55
+ FIM-Midtrain + SWE-Lego	35.00 (± 1.5)	32.70 (± 1.3)	33.85
Δ (ours vs. reproduced)	+3.20	+5.40	+4.30

203 pipeline; the cross-base-model comparison thus varies the post-training pipeline simultaneously, a
204 confound we discuss in Section 3.2.

205 3.2 Main Results on SWE Agent Benchmarks

206 Table 1 summarizes in-domain agent results.

207 **Consistent gains across model scales.** Holding the post-training pipeline fixed at R2E-Gym, FIM
208 mid-training improves SWE-Bench-Verified by +2.80 at 7B and +3.00 at 14B; the absolute gain is
209 essentially unchanged as the base model doubles in parameters. SWE-Bench-Lite shows the same
210 direction at both scales (+3.67/ +4.00). We read this as evidence that the structural prior is not
211 absorbed by larger pretrained models in the practical deployment range we evaluate.

212 **Transfer across post-training pipelines.** Replacing R2E-Gym with SWE-Smith on the same 7B
213 base yields +5.30 points on Verified, larger than the +2.80 under R2E-Gym; on Lite the SWE-Smith
214 pairing gains only +0.50, smaller than +3.67 for the R2E-Gym pairing. The two pipelines together
215 indicate that mid-training is not tuned to a single post-training data distribution, though the magnitude
216 of its benefit depends on the baseline pipeline it composes with.

217 **Transfer across base-model families.** Switching to Qwen3-8B and SWE-Lego, mid-training
218 improves Verified by +3.20 and Lite by +5.40, comparable to the Qwen2.5-Coder gains. This
219 comparison varies the post-training pipeline simultaneously with the base model (SWE-Lego is
220 required because Qwen3-8B’s 40,960-token context exceeds that of the R2E-Gym/SWE-Smith
221 scaffolds), so the cross-base-model result should be read as “not specific to the Qwen2.5-Coder
222 family under one alternative agentic pipeline.”

223 3.3 Capability Preservation and Cross-Domain Transfer

224 A natural concern with task-specialized post-training is that it erodes capabilities the base model
225 already had. We test this on six benchmarks outside SWE-Bench and—for τ -bench and BFCL—

Table 2: Capability preservation and cross-domain transfer at 14B with R2E-Gym. Bold marks the better trained variant per column (post-only vs. ours); the Instruct row is shown as a reference ceiling. All cells are percentages.

Setting	Non-agent coding			Agent OOD	Tool use		Avg
	LiveCode	OJBench	FSB-EN	Terminal	τ -bench	BFCL	
Instruct Model	37.20	5.20	53.80	0.00	5.70	23.20	20.85
+ R2E-Gym	24.10	2.80	47.72	2.41	3.40	15.80	16.04
+ FIM Mid-Train + R2E-Gym	35.20	4.74	48.25	3.66	7.30	18.20	19.56
Δ (vs. R2E-Gym only)	+11.10	+1.94	+0.53	+1.25	+3.90	+2.40	+3.52

outside the coding domain entirely. Because the controlled triple (instruct base vs. post-training only vs. mid-training + post-training) is the most expensive to run, we conduct it at 14B with R2E-Gym (Table 2).

Agentic post-training has a substantial hidden capability cost. R2E-Gym alone reduces every non-agent and tool-use benchmark relative to the Instruct ceiling: LiveCodeBench by 13.10, BFCL by 7.40, FullStackBench-EN by 6.08, τ -bench by 2.30, and OJBench by 2.40. Averaged across the six benchmarks the post-trained model loses 4.81 points relative to Instruct—the implicit cost paid for SWE-Bench gains, rarely highlighted in agent papers.

Mid-training largely closes the regression gap. Adding FIM mid-training before the same post-training restores LiveCodeBench by +11.10, OJBench by +1.94 (within 0.46 of the Instruct ceiling), and FullStackBench-EN by +0.53. The six-benchmark average rises from 16.04 to 19.56 (+3.52 over post-training only), while the SWE-Bench-Verified gain on the same base is preserved (Table 1). Mid-training therefore improves the cost–benefit profile of agentic post-training: in-domain target metrics improve and the bulk of off-distribution erosion is undone in the same training run.

Cross-domain transfer to non-coding tool use. τ -bench and BFCL contain no Python code-editing data, and our mid-training corpus contains no tool-use trajectories. Mid-training nevertheless improves τ -bench by +3.90 and BFCL by +2.40 over post-training alone, with a directionally consistent recovery on Terminal-Bench (+1.25). Because the mid-training data carries no tool-use signal, the only mechanism by which it can affect these benchmarks is by installing a structural prior at mid-training time that survives subsequent post-training—direct evidence for the function-call/tool-call isomorphism (Section 2.1).

3.4 Ablation Studies

We run two controlled ablations on Qwen2.5-Coder-7B with R2E-Gym post-training and a shared mid-training-free baseline. The first isolates the role of the chain-of-thought rationale; the second isolates the role of the function-aware selection pipeline. All selection variants in block (B) operate under a controlled budget—each variant selects exactly 200K single functions, with hard filters ($\text{LoC} \in [10, 200]$, dunder methods excluded) applied uniformly—so differences across rows reflect *which* functions are selected rather than *how many*. Block (C) varies mask granularity. Due to compute constraints, ablations run at 7B only.

FIM structure contributes independently of CoT distillation. Block (A) addresses a natural concern that improvements stem mainly from distilling reasoning. Removing the rationale entirely (*no CoT*) lifts the average by only +0.13 over the baseline (and +0.60 on Verified): in absolute terms the marginal signal from FIM structure alone is small, which we attribute to most of the underlying code already being seen by the base model during pretraining. The more informative comparison is against a frontier teacher. Replacing Gemini-3 with rationales generated by the model under training (*self-CoT*) reaches 14.85, recovering 1.68 of the 2.30-point gap to *Gemini-3 CoT*: high-quality external rationales help, but a sizable fraction of the gain is reproducible without a frontier teacher. The function-aware FIM objective is therefore effective even when paired with the model’s own rationales, and our recipe does not merely distill from a frontier teacher.

Function-selection algorithm is the dominant lever. Block (B) varies the selection algorithm with CoT and budget held fixed. *Random* masking establishes the floor at 13.95, a margin of +0.78 over no mid-training: even random function-level masking on top of the basic hard filters helps modestly.

Table 3: Ablations on the 7B model with R2E-Gym post-training. (A) rationale source. (B) function-selection algorithm. (C) mask granularity (single vs. multi-function groups). All blocks share the baseline (*w/o mid-train*) and a controlled 200K-target budget; CoT is fixed to Gemini-3-Flash in (B) and (C). Bold marks the best configuration per block; the 80%/15%/5% mixture in (C) is the recipe used in our main results (Table 1), which is trained on the full corpus rather than the 200K budget here.

Setting	SWE-Bench-Verified	SWE-Bench-Lite	Average
<i>(A) Chain-of-thought source (selection fixed to "full")</i>			
w/o mid-train (baseline)	15.00	11.33	13.17
+ FIM, no CoT	15.60	11.00	13.30
+ FIM, self-CoT (Qwen2.5-Coder-7B)	16.40	13.30	14.85
+ FIM, Gemini-3-Flash CoT	17.00	14.20	15.60
<i>(B) Function-selection algorithm (CoT fixed to Gemini-3-Flash)</i>			
w/o mid-train (baseline)	15.00	11.33	13.17
Random	15.30	12.60	13.95
Gemini-selected	16.40	13.70	15.05
PDG only	16.10	13.60	14.85
PDG + complexity ($\hat{H}$)	16.50	13.60	15.05
PDG + inferability ($\hat{I}$)	16.70	14.00	15.35
Full (PDG + $\hat{H}$ + $\hat{I}$)	17.00	14.20	15.60
<i>(C) Mask granularity (selection fixed to "full", CoT fixed to Gemini-3-Flash)</i>			
w/o mid-train (baseline)	15.00	11.33	13.17
Single-function only	17.00	14.20	15.60
85% single + 15% pair ($k = 2$)	17.20	14.60	15.90
95% single + 5% triple ($k = 3$)	17.00	14.40	15.70
80% single + 15% pair + 5% triple	17.40	14.80	16.10

Letting Gemini-3 choose which functions to mask (*Gemini-selected*) reaches 15.05 (+1.10 over Random), showing that frontier judgment on *which* function to mask is helpful but not by itself sufficient. Restricting the candidate pool to functions with at least one PDG neighbor (*PDG only*) reaches 14.85, between Random and Gemini-selected: structural connectivity alone contributes a small gain. Adding a scoring function on top of the PDG pool yields the next layer: *PDG* + $\hat{H}$ reaches 15.05 and *PDG* + $\hat{I}$ reaches 15.35, showing that intrinsic difficulty and contextual recoverability each contribute beyond the structural filter. Combining both via the harmonic-mean-like form (*Full*) reaches 15.60 (+0.25 over the better single-score variant), so $\hat{H}$ and $\hat{I}$ measure complementary properties and their combination is not redundant.

Mask granularity: pair masking helps, triples saturate. Block (C) mixes single-function targets with multi-function groups (Section 2.3.5). Adding 15% pair targets on top of single functions raises the average from 15.60 to 15.90 (+0.30); substituting 5% triple targets instead is essentially neutral (15.70). The full 80%/15%/5% mix achieves 16.10 (+0.50 over single-only), with the largest gain on Lite (+0.60). The pattern is consistent with the analysis in Section 4.2: training on cross-function dependencies disproportionately helps tasks whose gold patches span multiple functions, but the marginal return from larger groups diminishes as coupling becomes harder to maintain in the joint-masked context.

4 Analysis

The previous section established that FIM mid-training improves end-task metrics. We now ask *how* the resulting agent behaves differently and *where* along the trajectory the gains accrue. Both analyses use the 14B configuration with R2E-Gym, comparing the post-training-only baseline (*R2E-Gym*) against our recipe (*FIM-Midtrain* + *R2E-Gym*); statistics are run-means over three evaluation runs on SWE-Bench-Verified. The Lite analogue, the full action-type distribution, the no-patch mechanism, and a concrete trajectory contrast are deferred to Appendix D.

4.1 Recovery from Negative Observations

A trajectory contains a *negative observation* if any tool output matches a fixed set of error patterns (stack traces, “No replacement was performed”, shell errors, etc.; full list in Appendix B). The fraction of such trajectories is essentially identical across checkpoints (88.8% baseline vs. 91.8% ours), so the agents see comparable amounts of negative feedback. The *recovery rate*—the fraction of error-containing trajectories that nonetheless terminate with a passing patch—rises from 24.8% to 28.8% (+4.0 pp; +16% relative; Table 4). This is precisely the capability our framing predicts mid-training should support: continuing productively after an external return contradicts the model’s prior expectation. Mid-training also shifts the agent toward an iterate-and-verify policy, increasing edits per solved task from 3.3 to 7.4 and trajectory length from 15.1 to 23.6 steps; action-type breakdowns are in Appendix D.

Table 4: Headline trajectory metrics on SWE-Bench-Verified (14B, R2E-Gym), run-means over three runs. Full metrics in Appendix D.

Setting	Recovery rate (%)	Edits / solved	Steps / solved	Pass (%)
+ R2E-Gym	24.8	3.3	15.1	26.2
+ FIM-Midtrain + R2E-Gym	28.8	7.4	23.6	29.2

4.2 The Gain Concentrates on Multi-Function Reasoning

The most direct test of the isomorphism argument (Section 2.1) is whether mid-training preferentially helps tasks that require reasoning about cross-function dependencies inside a file. We stratify the 500 Verified tasks by the shape of the gold patch (Figure 4, Appendix D). On the 88 tasks whose gold patch modifies ≥ 2 functions within a single file, the baseline solves 13.6% on average and ours solves 22.7%, an absolute gain of +9.1 pp—more than $4\times$ the gain on the 341 single-function tasks (+2.1 pp). Per-instance head-to-head on this bucket shows ours uniquely solves about twice as many tasks as the baseline. Multi-file tasks ($n=71$) are not differentially helped, which we attribute to a granularity mismatch: our FIM operates within files (Appendix D). The slice where mid-training helps most is precisely the slice where the agent must follow control- and data-dependencies between caller-callee or sibling functions inside a file—the same structure exposed by function-aware FIM masking. A complementary outcome-distribution breakdown (Figure 5) shows the +15-task gain is driven by an order-of-magnitude reduction in *no-patch* failures alongside small reductions in localization errors; the FIM signal conditions the model to produce a non-empty span between prefix and suffix—a disposition that survives subsequent post-training.

5 Related Work

Mid-training and continued pretraining. A growing line of work identifies a distinct stage between large-scale pretraining and task-specific post-training, in which a model is further trained on a smaller, carefully curated corpus to install inductive biases that are hard to acquire from generic web text or downstream fine-tuning. MiniCPM [6] and OLMo [4] schedule such a stage to upweight high-quality or domain-relevant data near the end of pretraining, and Code-Llama [16] long-context adaptation can be read as a mid-training stage targeting context-length generalization. We adopt the same staging philosophy but apply it to a different end: installing an *agent-oriented* structural prior through function-aware FIM.

Fill-in-the-middle objectives. The fill-in-the-middle objective [1] is a standard ingredient in modern code LLM pretraining, with random-span variants used in Code-Llama [16], StarCoder [11], Qwen-Coder [7], and DeepSeek-Coder [5]. Our recipe extends and specializes prior FIM along three axes. First, masking targets are chosen at *function granularity* via program dependency graph analysis and a complexity–inferability double criterion, so the masked region corresponds to a unit of reasoning rather than an arbitrary span. Second, we embed an *explicit chain-of-thought* inside the FIM middle, supervising the model to produce a rationale before the code. Third, the objective lives in a *dedicated mid-training stage* rather than dissolved into pretraining.

Coding agent foundation models. Progress on coding agents has been driven by benchmarks such as SWE-Bench [10], scaffolds such as SWE-agent [21] and OpenHands [18], and trajectory-centric post-training pipelines such as SWE-Gym [14], r2e-gym [9], and SWE-Smith [22]. Our work is

complementary: we use r2e-gym and SWE-Smith unmodified on top of a mid-trained checkpoint, and the novelty lies one stage earlier—extracting a self-supervised signal from the structure already present in source code, which both raises in-domain agent metrics and mitigates the off-domain capability erosion that trajectory-only post-training otherwise inflicts.

Distillation from frontier models. Our default recipe uses Gemini-3-Preview to generate the chain-of-thought rationales embedded in each FIM middle span, placing it within the tradition of distilling capability from a strong teacher [13, 19, 3]. We isolate the contribution of FIM structure from CoT distillation through a controlled ablation in Section 3.4: removing the rationale still beats the no-mid-training baseline, and replacing the frontier teacher with self-generated rationales recovers a sizeable fraction of the gain, so the recipe is not distillation-bound.

6 Limitations and Discussion

We close by stating four limitations that scope our claims and flag directions for follow-up work. (i) *Python-only corpus and evaluation.* The mid-training corpus and the in-domain agent benchmarks (SWE-Bench-Verified, SWE-Bench-Lite) are exclusively Python; cross-language evidence comes only indirectly through FullStackBench-EN (Section 3.3), and whether the recipe transfers to Java, C++, or Rust ecosystems is left to future work. (ii) *Teacher dependency for CoT.* The default recipe relies on Gemini-3-Preview to synthesize rationales; the CoT-source ablation (Section 3.4) shows self-generated rationales recover most of the gain, but a fully open-source replication of our best configuration requires a comparably strong open teacher or acceptance of the residual gap. (iii) *Partial cross-base validation.* Our cross-base evidence comes from a single non-Qwen2.5-Coder configuration (Qwen3-8B with swe-lego), which simultaneously varies the post-training pipeline; the result should be read as “not specific to a single pretraining/post-training combination” rather than as a guarantee across all base-model families. (iv) *Modularity assumption.* Function-aware FIM presupposes meaningfully modular code; on monolithic scripts, generated code, or notebooks, the selection pipeline has fewer eligible targets and degrades to a smaller usable subset, a regime we do not study systematically here.

7 Conclusion

We started from the observation that a single step of a coding agent and a single function call site share the same four-part structure—*context*, *action*, *externally produced return*, *continuation*—making source code an internet-scale supply of agent-relevant signal. We turn this into function-aware FIM mid-training: a self-supervised stage that masks targets selected via program dependency graph analysis and a complexity–inferability double criterion, with chain-of-thought rationales embedded inside the FIM middle. Across three axes of robustness—model size (7B/14B), post-training pipeline (r2e-gym, SWE-Smith, swe-lego), and base-model family (Qwen2.5-Coder, Qwen3)—the recipe delivers consistent gains on coding-agent benchmarks, and the same Python-only corpus also transfers to non-coding tool-use benchmarks (τ -bench, BFCL) where it has no a priori reason to help. Future work includes extending function-aware selection to non-Python languages and composing mid-training with reinforcement-learning post-training.

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479 A Corpus Details

480 This appendix expands on the data-collection summary (Section 2.2). We report category coverage,
481 per-property statistics, and the license inventory of the 968 released repositories.

482 A.1 License Inventory

483 Table 6 summarizes the license distribution across the 968 released repositories. The corpus is
484 overwhelmingly permissive: $\sim 79\%$ of repositories carry a permissive license (MIT, Apache-2.0,
485 BSD-2/3, ISC, Boost, MIT-0, Unlicense, WTFPL); $\sim 5\%$ carry a copyleft license (GPL-2/3, AGPL-
486 3, LGPL-2.1/3, MPL-2.0, BSD-3 Clear); and $\sim 2\%$ carry Creative Commons licenses, with the

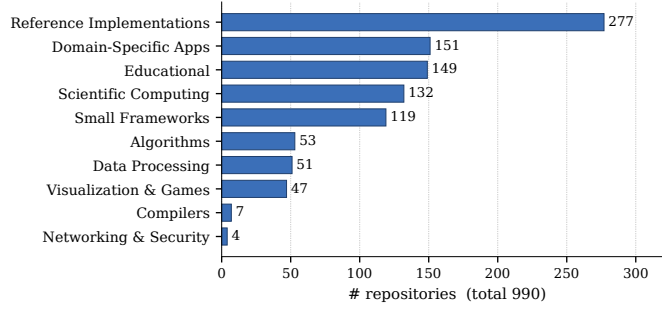


Figure 3: Distribution of the 968 source repositories across ten topic categories. The corpus is dominated by reference implementations, scientific computing, and small frameworks; compiler and networking/security tails are kept by design to maintain coverage diversity.

Table 5: Mid-training corpus statistics after decontamination and quality filtering. Token counts use the Qwen2.5-Coder tokenizer. Red entries are placeholders awaiting final replication runs.

Property	Value
Source GitHub repositories	968
Topic categories	10
Filtered self-contained Python files	$\approx 78\text{K}$
Total lines of Python code	$\approx 33\text{M}$
Mean LoC per file	≈ 428
Mid-training token budget	$\approx 2.6\text{B}$
Single-function FIM targets	$\approx 320\text{K}$ ($\approx 2.0\text{B}$ tokens)
Multi-function targets ($k=2$)	$\approx 60\text{K}$ ($\approx 0.4\text{B}$ tokens)
Multi-function targets ($k=3$)	$\approx 20\text{K}$ ($\approx 0.2\text{B}$ tokens)
Mean target LoC	≈ 34
Targets with Gemini-3 CoT	100%
SWE-Bench source-repo overlap	0
Commit cutoff	before earliest SWE-Bench base

487 remainder either lacking a declared license or unidentifiable by the crawler. We will release the
488 per-repository license alongside the repository list.

Table 6: License inventory of the 968-repository corpus. Counts are exact at release time; a small number of repositories with no declared or unidentifiable license are reported separately.

License	Count	License	Count
MIT License	TBD	GNU AGPL v3.0	TBD
Apache License 2.0	TBD	CC BY-SA 4.0	TBD
BSD 3-Clause	TBD	CC BY 4.0	TBD
BSD 2-Clause	TBD	CC0 1.0	TBD
GNU GPL v3.0	TBD	MPL 2.0	TBD
GNU GPL v2.0	TBD	GNU LGPL v2.1	TBD
ISC License	TBD	GNU LGPL v3.0	TBD
MIT-0	TBD	CC BY-NC 4.0	TBD
The Unlicense	TBD	BSD 3-Clause Clear	TBD
Boost Software License 1.0	TBD	Other / unidentifiable	TBD
Total: 968			

489 B Algorithmic Details

490 This appendix collects the full algorithmic details summarized in Section 2.3.

491 B.1 Single-Function FIM Target Selection

492 Algorithm 1 summarizes the per-file pipeline that produces single-function FIM targets. It composes
 493 the program dependency graph (Section 2.3.1), the complexity score $\hat{H}$ (Section 2.3.2), the inferability
 494 score $\hat{I}$ (Section 2.3.3), the difficulty penalty ρ (Appendix B.5), and the hard filters described below.

Algorithm 1 Single-function FIM target selection for one file.

Require: Source file s ; thresholds $\tau_d, \tau_{\text{FIM}}, \tau_{\hat{H}}$

```

1:  $T \leftarrow \text{ParseAST}(s)$ 
2:  $\mathcal{V}, \mathcal{E}_{\text{call}}, \mathcal{E}_{\text{sib}} \leftarrow \text{BuildPDG}(T)$  ▷ Section 2.3.1
3:  $\mathcal{C} \leftarrow \emptyset$ 
4: for each  $v \in \mathcal{V}$  do
5:   if  $v$  is a dunder method or  $\text{LoC}(v) \notin [10, 200]$  then
6:     continue
7:   end if
8:   Compute  $\hat{H}(v)$  and  $\hat{I}(v)$  ▷ Eqs. (1), (2)
9:   if  $\hat{H}(v) < \tau_{\hat{H}}$  then continue
10:  end if
11:   $\Delta(v) \leftarrow \max(0, \hat{H}(v) - \hat{I}(v)) / (\hat{H}(v) + \epsilon)$ 
12:   $\text{FIM}(v) \leftarrow \frac{\hat{H}(v) \hat{I}(v)}{\hat{H}(v) + \hat{I}(v) + \epsilon} \cdot \rho(\Delta(v))$ 
13:  if  $\text{FIM}(v) \geq \tau_{\text{FIM}}$  then
14:     $\mathcal{C} \leftarrow \mathcal{C} \cup \{v\}$ 
15:  end if
16: end for
17: return  $\mathcal{C}$  ▷ ranked by FIM descending

```

495 B.2 Program Dependency Graph: Call Resolution

496 For every `Call` node inside the body of $u \in \mathcal{V}$, we attempt to resolve its callee to some $v \in \mathcal{V}$, adding
 497 a directed edge $u \rightarrow v$ in $\mathcal{E}_{\text{call}}$. Resolution handles three Python idioms:

- 498 • **Direct module-level invocations:** `foo(...)` is resolved by qualified-name lookup in the
 499 same module.
- 500 • **Class instantiation:** `ClassName(...)` is resolved to `ClassName.__init__`.
- 501 • **Method calls:** `self.m(...)` or `cls.m(...)` inside class bodies is resolved within the
 502 enclosing class scope.

503 When a call cannot be resolved exactly we fall back to short-name matching against the qualified-
 504 name index, which recovers most cross-module references while admitting a controlled false-positive
 505 rate. Sibling edges $\mathcal{E}_{\text{sib}}$ are constructed by enumerating all pairs of methods belonging to the same
 506 class.

507 B.3 Complexity Score: Caps and Weights

508 The complexity score $\hat{H}(v)$ defined in Eq. (1) uses caps $(c_\ell, c_c, c_d) = (50, 10, 5)$ and weights
 509 $(w_\ell, w_c, w_d) = (0.4, 0.4, 0.2)$. The cap $c_\ell = 50$ on lines of code matches the median target length in
 510 our corpus; the cap $c_c = 10$ on McCabe complexity is roughly twice the median over our filtered
 511 functions; the cap $c_d = 5$ on nesting depth covers the tail of practical Python code. The cap value of
 512 2 in $\phi(x, c) = \min(x/c, 2)$ allows genuinely complex functions to score above the median without
 513 letting outliers dominate.

514 B.4 Inferability Score: Component Formulas

515 The five components of $\hat{I}(v)$ (Eq. 2) are defined below. Default mixing weights are $(\alpha, \beta, \gamma, \delta, \epsilon) =$
 516 $(0.30, 0.25, 0.20, 0.10, 0.15)$.

517 **Caller signal** C_{caller} . For each caller u of v , we score the specificity of its call sites by inspecting the
 518 syntactic form of each argument: literal constants are maximally informative, named variables less
 519 so, keyword arguments contribute proportionally, and a base score is awarded for the call site itself.
 520 The sum is normalized by an expected-callers constant.

521 **Callee signal** C_{callee} . The number of intra-file functions that v itself calls, normalized by a small
 522 expected-fan-out constant. A function that orchestrates known helpers is more recoverable than one
 523 calling only opaque externals.

524 **Signature signal** C_{sig} . A monotone aggregate of return-type annotation, parameter-type annotations,
 525 the descriptiveness of the function name (decomposed into snake-case parts), and the count of
 526 non-self parameters.

527 **Documentation signal** C_{doc} . Indicator of docstring presence. We do not score docstring informative-
 528 ness, both for robustness and because Section 2.4 generates a richer textual rationale.

529 **Class signal** C_{class} . For methods, the number of in-class siblings (normalized) plus a small bonus
 530 when an `__init__` is present in the same class.

531 B.5 Difficulty Δ , Penalty ρ , and Hard Filters

532 The penalty $\rho(\Delta(v))$ in Eq. (3) is built from the residual difficulty $\Delta(v)$, defined as the share of
 533 complexity that is not explained by context:

$$\Delta(v) = \frac{\max(0, \hat{H}(v) - \hat{I}(v))}{\hat{H}(v) + \epsilon} \in [0, 1]. \quad (4)$$

534 The penalty ρ is one-sided: targets with $\Delta(v) \leq \tau_d$ are left intact, while targets above the threshold
 535 are damped by a Gaussian factor:

$$\rho(\Delta) = \begin{cases} 1, & \Delta \leq \tau_d, \\ \exp\left(-\frac{(\Delta - \tau_d)^2}{2\sigma^2}\right), & \Delta > \tau_d. \end{cases} \quad (5)$$

536 We use $\tau_d = 0.5$ and $\sigma = 0.2$. The asymmetry reflects an asymmetry in the underlying objective:
 537 trivially easy targets are already eliminated by the hard filters below, so we do not need to addition-
 538 ally reward high inferability; conversely, targets that remain hard even with full context become
 539 unlearnable noise and must be down-weighted.

540 **Hard filters.** Independent of the smooth score, we discard (i) functions with $\text{LoC}(v) \notin [10, 200]$ to
 541 control training-instance length; (ii) dunder methods (`__init__`, `__repr__`, etc.), which serve as
 542 scaffolding rather than reasoning targets; (iii) functions with $\hat{H}(v) < 0.15$ to remove trivial cases;
 543 and (iv) functions with $\text{FIM}(v) < 0.08$. A file with no surviving target contributes nothing to the
 544 corpus.

545 B.6 Worked Example: Scoring `Calculator.total`

546 To make the numbers in Figure 2(b) reproducible, this subsection walks through every quantity that
 547 goes into $\hat{H}$, $\hat{I}$, and the final FIM score for the running example. The target function is:

```

def total(self) -> int:
    """Sum positive integer entries."""
    s, n = 0, 0
    for v in self.history:
        if is_int(v):
            if v > 0:
                s = add(s, v)
                n += 1
    if n == 0: return 0
    return s

```

549 **Lines of code (LoC = 10).** Following Python’s ast convention, $\text{LoC}(v) = \text{end_lineno} - (\text{lineno} -$
 550 $1)$, the number of source lines from the `def` keyword through the last statement, both inclusive.
 551 Counting the ten lines above (including the docstring) gives $\text{LoC} = 10$.

552 **Cyclomatic complexity (CC = 5).** The McCabe counter starts at 1 and adds 1 per branching node:
 553 If/IfExp, For/While, ExceptHandler, (values-1) per BoolOp, and one per generator inside a
 554 comprehension. For Calculator.total: base (+1), the for loop (+1), if is_int (+1), if v>0
 555 (+1), if n==0 (+1), totalling CC = 5. Tuple unpacking and n+=1 are not branches.

556 **Maximum nesting depth (D = 3).** Depth increments only on structural containers (If, For, While,
 557 AsyncFor, With, AsyncWith, Try). The deepest path in total is for → if is_int → if v>0,
 558 giving D = 3.

559 **Complexity score $\hat{H} = 0.40$.** Plugging into Eq. 1 with the default $(w_\ell, w_c, w_d) = (0.4, 0.4, 0.2)$
 560 and $(c_\ell, c_c, c_d) = (50, 10, 5)$:

$$\begin{aligned} w_\ell \phi(10, 50) &= 0.4 \cdot 0.20 = 0.08, & w_c \phi(5, 10) &= 0.4 \cdot 0.50 = 0.20, \\ w_d \phi(3, 5) &= 0.2 \cdot 0.60 = 0.12, & \hat{H}(\text{total}) &= 0.08 + 0.20 + 0.12 = 0.40. \end{aligned}$$

561 **Inferability score $\hat{I} = 0.48$.** (i) *Caller specificity.* The only in-file caller is Calculator.mean,
 562 which invokes self.total() with no arguments. The specificity routine assigns 0.50 for the call
 563 site, 0 for arguments and keywords, and a +0.10 found bonus, yielding sp = 0.60. Aggregating over
 564 callers and normalizing by 3, $C_{\text{caller}} = \min(0.60/3, 1.0) = 0.20$ and $\alpha C_{\text{caller}} = 0.06$. (ii) *Callee*
 565 *count.* total calls is_int and add in-file (self.history.append(v) does not count because
 566 append is not in $\mathcal{V}$), so $C_{\text{callee}} = \min(2/4, 1.0) = 0.50$ and $\beta C_{\text{callee}} = 0.13$. (iii) *Signature.* Sub-
 567 bonuses +0.30 for → int, +0.20 for the descriptive name total, 0 for parameter type annotations,
 568 0 for non-self parameters: $C_{\text{sig}} = 0.50$ and $\gamma C_{\text{sig}} = 0.10$. (iv) *Documentation.* A docstring is
 569 present, so $C_{\text{doc}} = 0.5$ and $\delta C_{\text{doc}} = 0.05$. (v) *Class context.* total sits among three siblings; the
 570 base score is $\min(3/5, 1.0) = 0.60$, plus a +0.30 bonus for the distinct __init__ of Calculator,
 571 giving $C_{\text{class}} = 0.90$ and $\varepsilon C_{\text{class}} = 0.14$. Summing: $\hat{I}(\text{total}) = 0.06 + 0.13 + 0.10 + 0.05 + 0.14 =$
 572 0.48.

573 **Final FIM score and selection.** $\Delta(\text{total}) = \max(0, -0.08)/0.40 = 0$, so $\rho(\Delta) = 1$ and
 574 $\text{FIM}(\text{total}) = (0.40 \cdot 0.48)/(0.40 + 0.48) \approx 0.22 \geq \tau = 0.20$, selecting it as a mask target. The four
 575 short helpers are removed before scoring by the LoC hard filter; Calculator.__init__ is removed
 576 by the dunder filter.

577 B.7 Multi-Function Group Score

578 **Group score.** For a group $G = \{v_1, \dots, v_k\}$ enumerated from one of eight topology patterns,

$$\text{FIM}(G) = \text{Coup}(G) \cdot \frac{\hat{H}(G) \hat{I}(G)}{\hat{H}(G) + \hat{I}(G) + \epsilon} \cdot \rho(\Delta(G)), \quad (6)$$

579 where $\hat{H}(G) = \frac{1}{k} \sum_v \hat{H}(v)$ averages individual complexities, $\Delta(G)$ is defined analogously to the
 580 single-function case, and the coupling term $\text{Coup}(G) \in [0, 1]$ is a normalized count of intra-group
 581 edges plus a shared-state ratio:

$$\text{Coup}(G) = w_c \frac{|E_G^{\text{call}}|}{k(k-1)} + w_s \frac{|E_G^{\text{sib}}|}{\binom{k}{2}} + w_{\text{st}} \overline{\text{Jacc}}_G, \quad (7)$$

582 with $\overline{\text{Jacc}}_G$ the mean Jaccard similarity over pairs in G between sets of accessed instance attributes
 583 (self.x). We use $(w_c, w_s, w_{\text{st}}) = (0.50, 0.20, 0.30)$. The coupling term enters multiplicatively
 584 because, unlike the single-function case, the value of a multi-function target derives specifically
 585 from cross-function dependencies; a weakly coupled group is indistinguishable from k independent
 586 samples.

587 **Group inferability.** $\hat{I}(G)$ is recomputed under the assumption that *all* members of G are masked
 588 simultaneously: caller, callee, and sibling contributions from inside G are excluded, and docstrings of
 589 group members contribute zero (they are part of the masked body). Signature information is retained
 590 because we do not mask function signatures. This re-computation prevents groups from receiving
 591 spurious credit for intra-group references that disappear under joint masking.

592 **Topology taxonomy.** A candidate group must form a connected subgraph in $\mathcal{E}_{\text{call}} \cup \mathcal{E}_{\text{sib}}$. We
 593 enumerate eight patterns:

- 594 • $k = 2$: *caller-callee* (direct $A \rightarrow B$ edge); *co-callee* (two callees of a common caller);
 595 *sibling-coupled* (same-class methods sharing ≥ 1 instance attribute); *mutual-call* ($A \rightleftharpoons B$).
- 596 • $k = 3$: *call-chain* ($A \rightarrow B \rightarrow C$); *hub* ($A \rightarrow B, A \rightarrow C$); *fan-in* ($B \rightarrow A, C \rightarrow A$); *class-triad*
 597 (three pairwise state-sharing methods of the same class).

598 **Group selection.** We score every candidate group emitted by the topology enumerator and discard
 599 those with insufficient coupling, complexity, or score (defaults: $\text{Coup}(G) \geq 0.20$; $\hat{H}(G) \geq 0.18$;
 600 $\text{FIM}(G) \geq 0.10$). Groups exceeding LoC ratios of 0.30 ($k = 2$) or 0.40 ($k = 3$) of the file are
 601 rejected. Remaining candidates are sorted by $\text{FIM}(G)$ and selected greedily under non-overlap: a
 602 function may belong to at most one selected group per file. Per-file caps (5 pairs, 3 triples) prevent
 603 any single file from dominating the corpus.

Algorithm 2 Multi-function group FIM target selection for one file.

Require: Source file s ; per-function scores $\hat{H}(v), \hat{I}(v)$; thresholds $\tau_{\text{coup}}^G, \tau_{\hat{H}}^G, \tau_{\text{FIM}}^G$; per-file caps
 $(N_2, N_3) = (5, 3)$

- 1: $\mathcal{V}, \mathcal{E}_{\text{call}}, \mathcal{E}_{\text{sib}} \leftarrow \text{BuildPDG}(\text{ParseAST}(s))$
- 2: $\mathcal{G} \leftarrow \emptyset$
- 3: **for** each topology pattern $P \in \{\text{caller-callee, co-callee, sibling-coupled, mutual-call, call-chain, hub, fan-in, class-triad}\}$ **do**
- 4: Enumerate all connected subgraphs $G \subseteq \mathcal{V}$ matching P in $\mathcal{E}_{\text{call}} \cup \mathcal{E}_{\text{sib}}$
- 5: **for** each candidate G **do**
- 6: **if** $\text{LoC_ratio}(G, s) > \theta_k$ **or** $\text{Coup}(G) < \tau_{\text{coup}}^G$ **or** $\hat{H}(G) < \tau_{\hat{H}}^G$ **then**
- 7: **continue**
- 8: **end if**
- 9: Recompute $\hat{I}(G)$ under joint masking ▷ exclude intra-group references
- 10: $\Delta(G) \leftarrow \max(0, \hat{H}(G) - \hat{I}(G)) / (\hat{H}(G) + \epsilon)$
- 11: $\text{FIM}(G) \leftarrow \text{Coup}(G) \cdot \frac{\hat{H}(G) \hat{I}(G)}{\hat{H}(G) + \hat{I}(G) + \epsilon} \cdot \rho(\Delta(G))$
- 12: **if** $\text{FIM}(G) \geq \tau_{\text{FIM}}^G$ **then**
- 13: $\mathcal{G} \leftarrow \mathcal{G} \cup \{(G, P, \text{FIM}(G))\}$
- 14: **end if**
- 15: **end for**
- 16: **end for**
- 17: Sort $\mathcal{G}$ by $\text{FIM}(G)$ descending
- 18: $\mathcal{S} \leftarrow \emptyset$
- 19: **for** each $(G, P, \text{FIM}(G)) \in \mathcal{G}$ **do**
- 20: **if** any $v \in G$ already covered by $\mathcal{S}$ **or** per-pattern cap (N_2 for $k=2$, N_3 for $k=3$) exceeded **then**
- 21: **continue**
- 22: **end if**
- 23: $\mathcal{S} \leftarrow \mathcal{S} \cup \{G\}$
- 24: **end for**
- 25: **return** $\mathcal{S}$

604 B.8 Worked Example: A caller-callee Pair

605 To illustrate the group score, consider a small file with two top-level functions:

```

def normalize(text: str) -> str:
    """Lowercase and strip punctuation."""
    out = []
    for ch in text:
        if ch.isalnum() or ch.isspace():
            out.append(ch.lower())
    return "".join(out)

def word_freq(doc: str) -> dict[str, int]:
    """Count tokens of 'doc' after normalization."""
    cleaned = normalize(doc)
    counts = {}
    for tok in cleaned.split():
        counts[tok] = counts.get(tok, 0) + 1
    return counts

```

Topology and pre-checks. The pair $G = \{\text{normalize}, \text{word_freq}\}$ matches the *caller-callee* pattern with one call edge $\text{word_freq} \rightarrow \text{normalize}$. They are top-level (no class), so the sibling component of Coup is zero and the shared-state component is zero (no `self._` accesses). The combined LoC is roughly 14 lines, well under the $\theta_2 = 0.30$ file-ratio cap.

Coupling. With one call edge in a pair, $|E_G^{\text{call}}|/(k(k-1)) = 1/2$. Plugging into Eq. 7 with $(w_c, w_s, w_{st}) = (0.50, 0.20, 0.30)$: $\text{Coup}(G) = 0.50 \cdot 0.5 + 0 + 0 = 0.25$, comfortably above the $\tau_{\text{coup}}^G = 0.20$ floor.

Group $\hat{H}$. Per-function complexities (computed as in Section 2.3.2) come out to roughly $\hat{H}(\text{normalize}) \approx 0.36$ (LoC=7, CC=3, D=2) and $\hat{H}(\text{word_freq}) \approx 0.30$ (LoC=7, CC=2, D=2), giving $\hat{H}(G) \approx 0.33$.

Group $\hat{I}$ under joint masking. Both functions are masked, so the call edge between them no longer contributes to C_{caller} or C_{callee} . Signature information is retained: both have descriptive names and full type annotations (strong C_{sig}); both have docstrings (small but nonzero C_{doc}). With no surviving caller/callee/class signal, the group inferability collapses to $\hat{I}(G) \approx 0.18$, dominated by the signature term.

Final group FIM. The harmonic-mean-like product $\hat{H}\hat{I}/(\hat{H} + \hat{I}) \approx (0.33 \cdot 0.18)/0.51 \approx 0.116$, $\Delta(G) = (0.33 - 0.18)/0.33 \approx 0.45 < \tau_d = 0.5$ so $\rho = 1$, and $\text{Coup}(G) = 0.25$: $\text{FIM}(G) \approx 0.25 \times 0.116 \approx 0.029$. This example sits below the $\tau_{\text{FIM}}^G = 0.10$ floor and would therefore *not* be selected—the toy module is too small for the joint-masking inferability to remain large enough. Real corpus pairs that pass the threshold typically come from class-method pairs with non-trivial sibling and shared-state coupling, where $\hat{I}(G)$ retains substantial C_{class} contribution even under joint masking. The walkthrough nonetheless shows the mechanism: joint masking strictly reduces $\hat{I}$ relative to the single-function average, and the coupling factor must compensate.

B.9 Chain-of-Thought Prompt Template

The chain-of-thought rationale embedded inside each FIM middle span (Section 2.4) is generated by Gemini-3-Preview using the following prompt skeleton:

```

[System] You are an expert software engineer.

[User] Below is a Python source file with one function body redacted (shown as <MASKED>).

<file with target body redacted>

The redacted function has the following signature and (if present) docstring:

<signature + docstring>

The original function body was:

<original body>

Write a concise step-by-step rationale (in natural language) that explains how the body's
behavior could be recovered from the surrounding file—for example, by referencing call
sites, helper functions, and class-level state—without simply restating the code.

```

643 Rationales are constrained to at most 300 tokens and are filtered by length and basic format checks
644 before training. We use greedy decoding; retry logic and quality filters will be released with the
645 corpus.

646 B.10 Negative-Observation Patterns

647 The recovery-rate analysis (Section 4.1) flags a tool output as a *negative observation* when
648 it matches any of the following patterns: Python stack-trace prefixes (Traceback (most
649 recent call last):); error class names (SyntaxError, IndentationError, NameError,
650 ImportError, AttributeError, TypeError, ValueError); shell error markers (command
651 not found, permission denied, No such file or directory); and harness-specific failure
652 strings (No replacement was performed, Patch did not apply, tests failed). Patterns
653 are matched case-insensitively.

654 C Training Hyperparameters

655 **Stub: hyperparameters and token budgets to be finalized at camera-ready time.** We list the
656 categories of hyperparameters that this appendix will cover:

- 657 • *Mid-training schedule*: number of epochs over the FIM corpus, learning rate (peak and
658 warm-up steps), weight decay, gradient-clip norm, optimizer, and batch size for each base
659 model (Qwen2.5-Coder-7B/14B, Qwen3-8B).
- 660 • *Sequence packing*: maximum sequence length per base model (matching native pretraining
661 context), sample-packing strategy, attention masking across packed boundaries.
- 662 • *FIM tokenization*: native FIM sentinels per model family, rationale-vs.-body separator,
663 position of the rationale inside the middle span.
- 664 • *Post-training*: post-training pipeline configuration (R2E-Gym, SWE-Smith, SWE-Lego),
665 number of post-training epochs (note the SWE-Lego 2-epoch deviation discussed in Sec-
666 tion 3.1), learning-rate schedule, and seed.
- 667 • *Evaluation*: agent harness configuration per pipeline (R2E-Gym scaffold built on Open-
668 Hands; SWE-agent for SWE-Smith; OpenHands for SWE-Lego), maximum turn limits,
669 decoding temperature, per-step timeout, and seed values.

670 D Extended Behavioral Analysis (SWE-Bench-Verified)

671 This appendix collects the trajectory-level material that supplements the headline analysis in Section 4.
672 Unless stated otherwise, statistics are run-means over three independent evaluation runs of each
673 checkpoint on SWE-Bench-Verified (500 instances per run).

674 D.1 Pass Rate by Gold-Patch Shape

675 Figure 4 reports the pass-rate stratification used in Section 4.2: the largest gain (+9.1 pp) is on
676 multi-function single-file tasks ($n=88$), more than $4\times$ the +2.1 pp gain on single-function tasks
677 ($n=341$); multi-file tasks ($n=71$) remain hard for both checkpoints ($\sim 11.3\%$ each).

678 D.2 Full Trajectory Metrics

679 Table 7 reports the full trajectory metrics summarized in Table 4, including unsolved-task breakdowns
680 and output/context summaries.

681 D.3 Iterate-and-Verify: Action-Type Distribution

682 Mid-training shifts the action-type distribution toward an iterate-and-verify policy. The share of
683 search actions falls from 15.3% to 11.0%, while the share of `execute_bash` actions rises from
684 19.5% to 24.6%: the agent spends a smaller fraction of its budget on passive repository lookup
685 and a larger fraction on actively running scripts and tests, then refining its edit. The cost is more

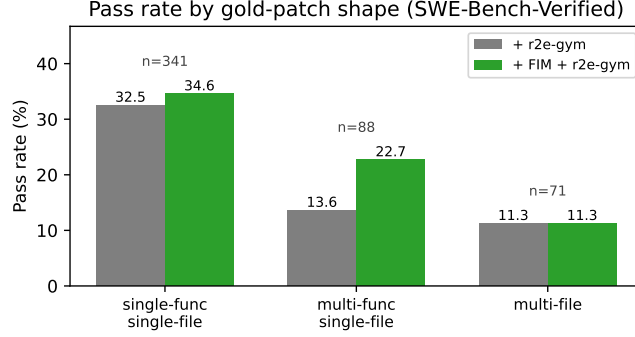


Figure 4: Pass rate on SWE-Bench-Verified (14B + R2E-Gym) stratified by gold-patch shape, run-means over three evaluation runs per checkpoint.

Table 7: Trajectory-level behavioral metrics on SWE-Bench-Verified (14B, R2E-Gym), run-means over three evaluation runs of each checkpoint. Steps, edits, and errors-encountered are means per task. “Empty patch” is the fraction of *failed* trajectories whose final `output_patch` is empty; “Loc. correct” is the fraction of all trajectories that locate at least one file overlapping the gold patch; “Ctx @ success” is the mean prompt context length on solved trajectories.

Setting	Steps / task		Edits / task		Errors enc.		Recovery rate (%)
	solved	unsolved	solved	unsolved	solved	unsolved	
+ R2E-Gym	15.1	22.4	3.3	5.4	3.5	5.8	24.8
+ FIM-Midtrain + R2E-Gym	23.6	32.7	7.4	10.9	6.2	8.1	28.8
<i>Output and context summaries</i>							
	Empty patch (% failed)		Loc. correct (% all)		Ctx @ success (tok)		Pass (%)
+ R2E-Gym	3.0		70.6		11,826		26.2
+ FIM-Midtrain + R2E-Gym	0.3		73.4		16,397		29.2

steps—ours hits the step cap on $\sim 45\%$ of trajectories vs. 7% for the baseline—but this cost is paid on the unsolved tail; on the solved set the additional steps translate into more correct patches.

D.4 Failure-Mode Breakdown

We label every failed trajectory using a deterministic patch-quality cascade: *no-patch* (output diff is empty); *localization error* (non-empty patch but zero file overlap with the gold patch); *patch error* (file overlap with gold patch but tests fail). Figure 5 reports the outcome distribution: across the three runs, mid-training cuts no-patch failures by an order of magnitude ($\sim 11 \rightarrow \sim 1$ trajectories per run), trims localization errors slightly ($\sim 131 \rightarrow \sim 126$), and leaves the patch-error count essentially flat ($\sim 227 \rightarrow \sim 227$). The +15-task gain in solved count is therefore primarily driven by the no-patch mode collapse.

D.5 No-Patch Mode: Mechanism

The baseline no-patch failures recovered by mid-training are distributed across all task buckets and are not concentrated in any single repository. In every recovered case the baseline emitted `<finish>` without applying any `str_replace`, while ours applied at least one edit. We interpret this as a direct consequence of the FIM training signal: under FIM the model is always conditioned to produce a non-empty token span between the prefix and suffix, and this disposition survives the post-training pipeline.

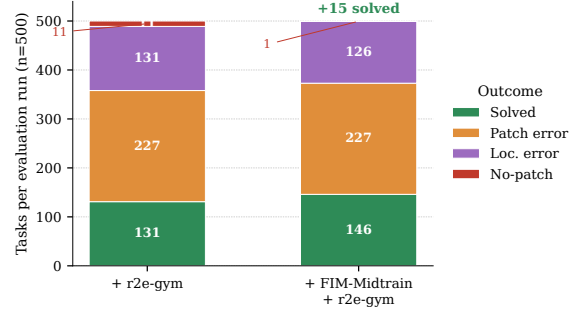


Figure 5: Outcome distribution per evaluation run on SWE-Bench-Verified (14B, R2E-Gym), averaged over three runs.

D.6 Multi-File Tasks Are Not Differentially Helped

On the 71 Verified tasks whose gold patch spans ≥ 2 files, both checkpoints solve $\sim 11.3\%$ on average and per-instance head-to-head on this bucket is balanced. We attribute this to a granularity mismatch: our function-aware FIM mid-training operates at the function level within files, so cross-file coordination is not directly trained for. Extending the selection algorithm to cross-file function pairs is a natural follow-up.

D.7 A Concrete Contrast

The shift in termination behavior is clearest on cases where the baseline quits prematurely. On `scikit-learn-26323`, for instance, the baseline trajectory runs for a single step in every run, immediately emits `<finish>` with an empty edit and the wrong target file, and is scored as a failure; on the same task our agent runs ~ 41 steps, applies on the order of 20 `str_replace` operations, observes multiple negative tool returns, and recovers to a passing patch. Pooling across the three runs, on the order of ~ 16 of the ~ 55 tasks that ours uniquely solves on Verified have this signature: the baseline emits `<finish>` with zero file overlap with the gold patch (often before any negative observation has even arrived), while ours iterates past intermediate signals and converges on a correct edit. Mid-training does not enable a categorically new capability on these instances; it changes the agent’s stopping policy.

E Behavioral Analysis on SWE-Bench-Lite

This appendix replicates the Verified analysis on the SWE-Bench-Lite test split (300 instances). The same trajectory-parsing pipeline and failure-mode taxonomy are used; statistics are run-means over three independent evaluation runs of each checkpoint.

The qualitative trends from Verified hold on Lite: mid-training raises the recovery rate from 18.8% to 22.1%, eliminates the no-patch failure mode entirely (baseline averages ~ 8 no-patch failures per run, ours averages 0), and increases edit iteration on solved trajectories from 4.4 to 7.8 `str_replace` operations per task. The *multi-function* concentration of gain documented on Verified is not visible on Lite because Lite contains no multi-file tasks and only 54 multi-function single-file tasks; the bulk of Lite is single-function single-file tasks, on which the gain is +4.9 pp (19.5% $\rightarrow$ 24.4%).

Failure-mode counts on Lite move in the same direction as on Verified: the no-patch mode is eliminated ($\sim 8 \rightarrow 0$ trajectories per run), localization errors fall by about 11, and patch errors rise by about 7, reflecting the same iterate-and-verify shift.

Per-instance head-to-head. Aggregating per-task outcomes across the three runs (run-majority vote per checkpoint), the differential favors ours on Lite by a $\sim 1.8\times$ ratio overall and by a $\sim 2.1\times$ ratio on the single-function single-file bucket. On the multi-function single-file bucket the differential is balanced, consistent with the small absolute number of such tasks in Lite.

Table 8: Trajectory-level behavioral metrics on SWE-Bench-Lite (14B, R2E-Gym), run-means over three evaluation runs of each checkpoint. Definitions match Table 7.

Setting	Steps / task		Edits / task		Errors enc.		Recovery rate (%)
	solved	unsolved	solved	unsolved	solved	unsolved	
+ R2E-Gym	17.5	21.3	4.4	5.2	4.8	5.4	18.8
+ FIM-Midtrain + R2E-Gym	24.6	31.7	7.8	11.1	5.7	7.5	22.1
<i>Output and context summaries</i>							
	Empty patch (% failed)		Loc. correct (% all)		Ctx @ success (tok)		Pass (%)
+ R2E-Gym	3.3		65.0		13,958		18.0
+ FIM-Midtrain + R2E-Gym	0.0		71.0		18,104		22.0

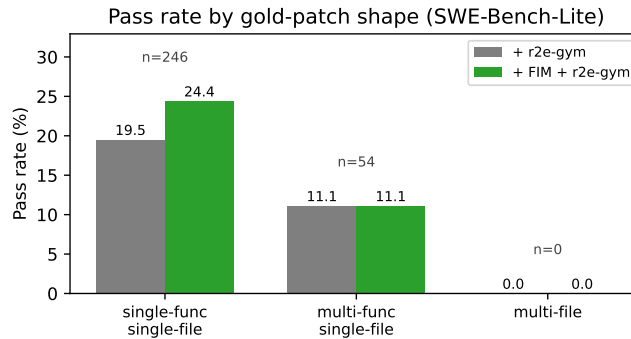


Figure 6: Pass rate on SWE-Bench-Lite stratified by gold-patch shape, averaged over three evaluation runs per checkpoint. Lite contains no multi-file tasks; the multi-function single-file bucket is small ($n=54$) and shows no gain on this slice, with the +4.0 pp end-task improvement coming entirely from the single-function single-file bucket ($n=246$, +4.9 pp).

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